RESEARCH PROPOSAL

Title:

MatriSuraksha: An AI-Driven Referral Decision Support System for Safer Childbirth in West Bengal

Principal Investigator (PI):

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ABSTRACT:

Maternal and neonatal health remains a significant public health concern in West Bengal, where despite recent improvements, the Maternal Mortality Ratio (MMR) of 105 per 100,000 live births and Neonatal Mortality Rate (NMR) above 12 per 1,000 live births continue to exceed national and SDG targets. Key challenges include late identification of high-risk pregnancies and delays in appropriate referrals from Primary Health Care level facilities (PHCs), especially in rural and underserved regions. The proposed "MatriSuraksha" project seeks to address these gaps by developing an AI-driven predictive analytics tool integrated with a Clinical Decision Support System (CDSS) to support PHC staff in identifying risks and making timely referral decisions during labour. This 24-month prospective observational study will be conducted across 12 PHCs in Dakshin Dinajpur district, covering diverse socio-economic and geographic profiles. A minimum of 1,200 pregnant women admitted for pregnancy-related events will be enrolled and followed through postpartum. Data will be collected using structured digital tools from medical records, labour room logs, and follow-ups. Machine learning models (e.g., Random Forest, XGBoost, etc.) will be developed using antenatal and intrapartum data to predict adverse maternal and neonatal outcomes such as eclampsia, obstructed labour, birth asphyxia, and neonatal mortality. The key deliverables include (1) a risk prediction module, (2) a clinical decision support engine, and (3) a user-friendly digital platform for PHC level officials. The project emphasizes ethical

compliance, data protection, and stakeholder collaboration with the state health department, frontline workers, and technology partners. MatriSuraksha aims to reduce delays in maternal care, improve outcomes, and strengthen India's progress toward SDG 3.1 and 3.2 and the National Digital Health Mission by integrating data-driven decision-making into grassroots maternal healthcare.

BACKGROUND:

Maternal and neonatal health remains a critical public health priority in India, particularly in states like West Bengal, where maternal mortality ratio (MMR) and neonatal mortality rate (NMR) continue to be influenced by socioeconomic disparities, rural-urban divides, and infrastructural limitations in the health system.[1] Despite decline in MMR and NMR in West Bengal, challenges persist in ensuring early identification and effective management of high-risk pregnancies, especially at the level of Primary Health Care (PHCs), where frontline health workers often lack real-time tools for risk stratification and timely referral decisions.[2] According to the latest Sample Registration System (SRS) Special Bulletin on Maternal Mortality in India (June 2025), West Bengal reports a Maternal Mortality Ratio (MMR) of 105 per 100,000 live births—higher than the national average of 88. While India's Neonatal Mortality Rate (NMR) has declined from 26 per 1,000 live births in 2014 to 19 in 2021, West Bengal's NMR remains above 12 per 1,000 live births, surpassing the SDG target of ≤12 by 2030. These figures underscore persistent gaps in antenatal surveillance, intrapartum monitoring, and postpartum care.[3, 4] Many maternal and neonatal deaths occur due to preventable causes such as hypertensive disorders, obstructed labour, infections, and delays in seeking or receiving appropriate care—widely conceptualized as the "three delays" model.[5] These delays—delay in recognizing danger signs, reaching appropriate health facilities, and receiving quality care—are particularly pronounced in rural and underserved populations. In this context, there is an urgent need for innovative, context-specific technological solutions that can augment existing health systems. The proposed "**MatriSuraksha**" project aims to leverage artificial intelligence (AI) and predictive analytics to support frontline healthcare providers by enabling early detection of maternal risk factors, stratification of case severity, and evidence-based referral decision-making. This tool has the potential to bridge the critical information and action gaps in maternal health service delivery, improve outcomes for mothers and newborns, and contribute directly to India's targets under the Sustainable Development Goals (SDG 3.1 and 3.2) and National Digital Health Mission.

OBJECTIVE:

To design and develop a predictive analytics tool integrated with a clinical decision support system (CDSS) aimed at improving maternal and neonatal outcomes by enabling early risk prediction and timely referrals during labour from PHCs to higher referral centres in West Bengal.

METHODOLOGY:

Study Design:

This will be a descriptive observational study with a technology development and validation component. The study will collect maternal and neonatal data from the Primary Health Care (PHCs) level units to develop, train, and validate the Artificial Intelligence (AI)-based predictive analytics and clinical decision support system (CDSS) for **timely referral to higher center**.

Study Setting:

The study will be conducted across the 12 (Twelve) Primary Level Health Care units (PLHCU), i.e. Primary Health Center/Block Primary Health Center/Rural Hospital in Dakshin Dinajpur district of West Bengal. To ensure comprehensive representation and meaningful insights, all the blocks will be selected to reflect diversity across key socio-economic, demographic, and geographic dimensions. This includes variation in rural–urban settings, tribal and non-tribal populations, levels of maternal and child health indicators, healthcare infrastructure, and socio-economic status.

Study Population:

Inclusion Criteria:

- Pregnant women admitted at the 12 PLHCU for any labor events.
- Willingness to provide informed consent and participate in follow-up till postpartum.

Exclusion Criteria:

- admitted for false labor or reasons unrelated to active labor, Unwilling or unable to provide informed consent or follow-up

Sample Size:

To ensure robust model development and validation, the project will aim to recruit at least 1200 pregnant women admitted at the 12 PLHCU at the Dakshin Dinajpur for any pregnancy related states or events over an 18-month study period. The sample size is calculated based on the approach described by Prof. Richard D Riley et. al. for developing a clinical prediction model using at least 40 samples per variable rule.[6] This sample size will provide sufficient power for predictive modeling and testing algorithm sensitivity, specificity, and area under the curve (AUC).

Sampling Technique:

All the pregnant women admitted at all PLHCU for any pregnancy related states or events during the study period will be consecutively enrolled.

Data Collection plan:

By Trained Field Data Collection Agents using Digital platform

Project Implementation and Data Collection Plan:

1. Tools and Instruments:

Structured Case Record Forms (CRFs) and digital data entry interface (tablet/mobile-based app). This will be finalized after a Pilot Study.

Standardized formats for:

- Antenatal data (as per last visit record): Age, Gravida, Parity, Previous LUCS history, Pulse, BP, Oedema, jaundice, Hb, Weight, Blood Grouping and Rh Typing, Urine albumin, Urine sugar, fetoplacental ultrasound findings, any comorbidities (e.g., anemia, gestational diabetes, Heart Disease, Tuberculosis, any other), Obstetric complications in previous pregnancy (if any),
- Intrapartum data: date & time of admission, duration and stage of labour, vital parameters, fetal heart rate monitoring, vaginal examinations.
- Referral Data: referred or nor, date and time of referral, reason for referral, place of referral
- Pregnancy outcomes: mode of delivery, maternal complications, neonatal status (birth weight, still birth, admission to SNCU/NICU, neonatal death if any)

2. Data Sources:

- Maternal and Child Health Protection card, labour room register, Partographs, referral register, and relevant logbooks.
- Interviews with Medical Officer/nurses and follow-up via home visits or phone calls to track maternal and neonatal outcomes.

3. Data Collection Phases:

- Phase 1 (Months 1–6): Baseline data collection, stakeholder consultations, Groundwork and Pilot Testing, finalize Case Record Form, training of the field staff, data collection tool development.
- Phase 2 (Months 7–18): Prospective data collection, follow-up till 42 days postpartum, tool iteration.
- Phase 3 (Months 19–24): ML based Decision support tool development and validation, refinement, and usability testing.

Data Analysis tools and Data Security:

- Python programmimg will be used for data analysis and AI-ML modeling and developing decision support tools and framework. Anonymized data will be used, and model will be developed locally.

Outcome Measures:

Primary Outcomes:

- Maternal events (e.g., Mode of delivery, eclampsia, postpartum hemorrhage, maternal mortality, any other adverse maternal outcome).
- Neonatal events (e.g., birth asphyxia, admission in SNCU/NICU, neonatal mortality).

Secondary Outcomes:

- Accuracy of AI-ML model in predicting complications (Sensitivity, Specificity, ROC-AUC).
- Timeliness and appropriateness of referral decisions.

Data Analysis Plan:

- Python programmimg will be used for data analysis and AI-ML modeling and developing decision support tools and framework. Anonymized data will be used, and model will be developed locally.
- Descriptive statistics for baseline characteristics in terms of central tendencies, deviations, association and correlations and visual representations.
- Supervised Machine learning models (e.g., Random Forest, XGBoost, Logistic Regression, etc.) for risk prediction using antenatal and intrapartum data.
- Evaluation metrics: ROC-AUC, Precision, Recall, F1 Score, Confusion Matrix.

Ethical Considerations:

- Institutional Ethics Committee (IEC) approval will be obtained.
- No new interventions will be applied during the study, only the available information will be collected.
- Informed consent will be taken from the participants. Depending on the circumstances, informed consent from the accompanying relative of the participants will be taken to use anonymous data for research purposes.
- Data will be anonymized and securely stored in compliance with data protection norms of Govt. of India.

Expected Outcomes and benefits:

- Improved data driven automated early detection of high-risk pregnancies and complications.
- Increased efficiency and appropriateness of referrals during Labour
- Reduced maternal and neonatal morbidity and mortality.
- Strengthened capacity of the staff at Primary Healthcare level in decision-making and emergency management during Labour.

Deliverables at the end of the project:

1. A **predictive analytics module** using historical antenatal and intrapartum data to identify high-risk pregnancies and complications (e.g., preeclampsia, obstructed labour, fetal distress).
2. A **decision support engine** to assist PLHCU staff (nurses, medical officers) in making timely referral decisions based on risk scores, symptoms, and labour

progression.
3. A **web-friendly digital platform** for usability by frontline health workers.

STAKEHOLDERS AND COLLABORATORS:

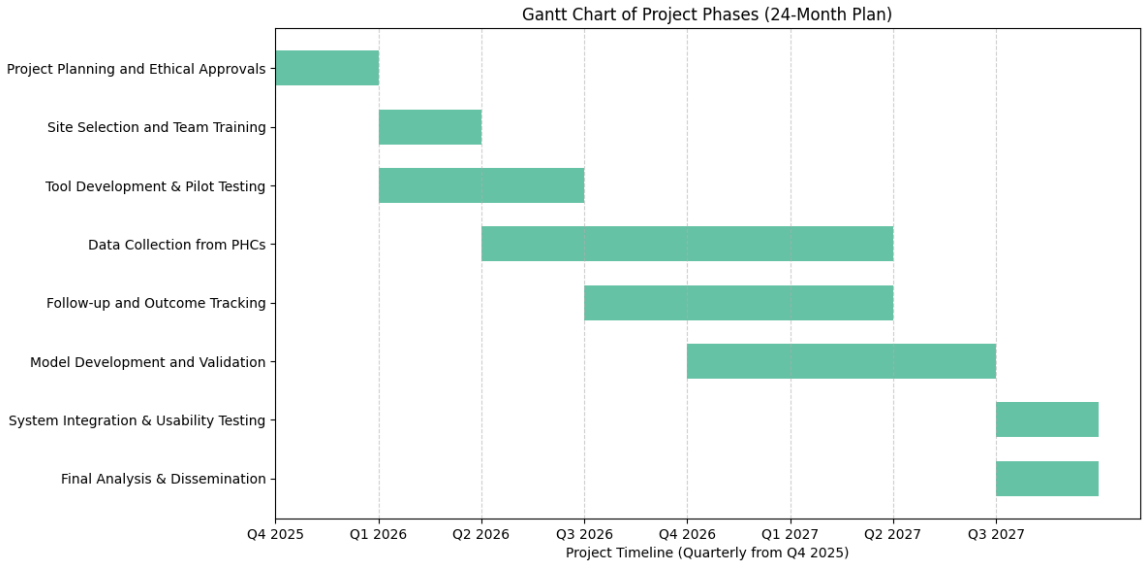
- Department of Health and Family Welfare, Government of West Bengal
- Public Health Researchers and Data Scientists
- District, Sub-Division and Block Health Administrators
- PHC Medical Officers, Nurses, ANMs, and ASHAs
- Technology Development Partners

CONCLUSION:

The **"MatriSuraksha"** initiative has the potential to revolutionize maternal healthcare at the grassroots level by leveraging data-driven insights and digital decision support to ensure timely and effective referrals. This will significantly contribute to achieving Sustainable Development Goals (SDG 3.1 and 3.2) in maternal and child health in West Bengal.

MatriSuraksha: Project Timeline & Estimated Budget

1. Gantt Chart / Project Timeline (24 Months) (Assuming Start: Q4 2025: Oct–Dec 2025)



2. Budget Estimation Framework (in INR Lakhs)

Budget Head	Amount (INR Lakhs)	Remarks
Personnel (Field Investigators, Data Managers, Coordinator)	6.00	4 part-time staff for data collection
Technology Development (Computers, AI Tool, CDSS Platform)	4.00	Model development, software engineering
Data Collection (travel, materials, logistics)	1.00	Tablets, printing, transportation
Training and Capacity Building	0.50	PHC staff and project team
Monitoring and Evaluation	0.50	IEC and external reviewers
Dissemination & Reporting	2.50	Workshops, publications
Overhead and Contingency	0.50	Unforeseen expenses

Total Estimated Budget: ₹15.00 Lakhs

Note: Seed Money of Rs. 50000.00 (Fifty Thousand) will be required for the initial expense to cover pilot project

References:

1. Ram U, Ramesh BM, Blanchard AK, Scott K, Kumar P, Agrawal R, Washington R, Bhushan H (2024) A tale of two exemplars: the maternal and newborn mortality transitions of two state clusters in India. BMJ Glob Health 9:e011413

2. Bango M, Ghosh S (2023) Reducing infant and child mortality: assessing the social inclusiveness of child health care policies and programmes in three states of India. BMC Public Health 23:1149

3. India - SAMPLE REGISTRATION SYSTEM (SRS)-SPECIAL BULLETIN ON MATERNAL MORTALITY IN INDIA 2020-22. https://censusindia.gov.in/nada/index.php/catalog/45569. Accessed 16 Jun 2025

4. India witnesses a steady downward trend in maternal and child mortality towards achievement of SDG 2030 targets. https://www.pib.gov.in/www.pib.gov.in/Pressreleaseshare.aspx?PRID=2128024. Accessed 16 Jun 2025

5. Actis Danna V, Bedwell C, Wakasiaka S, Lavender T (2020) Utility of the three-delays model and its potential for supporting a solution-based approach to accessing intrapartum care in low- and middle-income countries. A qualitative evidence synthesis. Glob Health Action 13:1819052

6. Riley RD, Ensor J, Snell KIE, Harrell FE, Martin GP, Reitsma JB, Moons KGM, Collins G, Van Smeden M (2020) Calculating the sample size required for developing a clinical prediction model. BMJ m441

PARTICIPANTS' CONSENT FORM

I, Ms/Mrs.....

W/o or d/o....., Aged……..

Resident of....., PIN

, do hereby declare that I have been informed about the study: "**MatriSuraksha: An AI-Driven Referral Decision Support System for Safer Childbirth in West Bengal"** at this hospital. I give voluntary consent to participate in the study. The nature and purpose of the study and its potential risks / benefits and expected duration of the study, and other relevant details of the study have been explained to me in detail. At any time during its course, my refusal to continue however is not going to affect my right to receive treatment from the department.
I do hereby declare that I shall provide clinical history, undergo clinical examination, allow collection of necessary clinical material and other relevant data, accept necessary treatment and come for follow up. I understand that my identity will not be revealed, and confidentiality of the information will be maintained.
I, do hereby agree to take part in this study.

Name of the declarant:

Signature of declarant:

Place:

Date:

Name of investigating officer:

Signature of investigating officer:

Place:

Dated:

प्रतिभागियों की सहमति प्रपत्र

मैं, श्रीमती .....

पुत्री/पत्नी ....., आयु .....

निवासी ....., पिन .....

यह घोषणा करता/करती हूँ कि मुझे इस अध्ययन के बारे में सूचित किया गया है: **"मातृसुरक्षा: सुरक्षित प्रसव हेतु पश्चिम बंगाल में एआई-संचालित रेफरल निर्णय सहयोग प्रणाली"**, जो इस संस्थान में किया जा रहा है।

मैं स्वेच्छा से इस अध्ययन में भाग लेने के लिए सहमति देता/देती हूँ। मुझे अध्ययन की प्रकृति और उद्देश्य, इसके संभावित लाभ/हानि, अपेक्षित अवधि तथा अन्य प्रासंगिक विवरणों के बारे में विस्तार से समझाया गया है। अध्ययन के दौरान किसी भी समय मेरा भाग न लेने का निर्णय, विभाग से उपचार प्राप्त करने के मेरे अधिकार को प्रभावित नहीं करेगा।

मैं यह भी घोषणा करता/करती हूँ कि मैं अपना चिकित्सकीय इतिहास प्रदान करूँगा/करूँगी, चिकित्सकीय परीक्षण कराऊँगा/कराऊँगी, आवश्यक नैदानिक सामग्री तथा अन्य प्रासंगिक आँकड़े उपलब्ध कराऊँगा/कराऊँगी, आवश्यक उपचार स्वीकार करूँगा/करूँगी तथा अनुवर्ती जाँच (फॉलो-अप) के लिए आऊँगा/आऊँगी। मुझे यह ज्ञात है कि मेरी पहचान उजागर नहीं की जाएगी तथा दी गई जानकारी को गोपनीय रखा जाएगा।

अतः मैं इस अध्ययन में भाग लेने के लिए सहमत हूँ।

घोषणाकर्ता का नाम:
घोषणाकर्ता के हस्ताक्षर:
स्थान:
दिनांक:

अन्वेषक अधिकारी का नाम:
अन्वेषक अधिकारी के हस्ताक्षर:
स्थान:
दिनांक:

অংশগ্রহণকারীদের সম্মতি পত্র

আমি, শ্রীমতী/ সুশ্রী .....

পিতা/মাতা/স্বামী/স্ত্রী ....., বয়স
.....

বাসিন্দা .....,
পিন .....

আমি ঘোষণা করছি যে আমাকে নিম্নলিখিত গবেষণা সম্বন্ধে অবহিত করা হয়েছে: **"মাতৃসুরক্ষা: নিরাপদ প্রসবের জন্য পশ্চিমবঙ্গে এআই-চালিত রেফারেল সিদ্ধান্ত সহায়ক ব্যবস্থা"**, যা এই প্রতিষ্ঠানে পরিচালিত হচ্ছে।

আমি স্বেচ্ছায় এই গবেষণায় অংশগ্রহণ করার জন্য সম্মতি দিচ্ছি। গবেষণার প্রকৃতি ও উদ্দেশ্য, এর সম্ভাব্য ঝুঁকি/লাভ, সম্ভাব্য সময়কাল এবং অন্যান্য প্রাসঙ্গিক বিষয় আমাকে বিস্তারিতভাবে জানানো হয়েছে। গবেষণার সময় যে কোনো মুহূর্তে অংশগ্রহণ প্রত্যাখ্যান করলে তা আমার চিকিৎসা পাওয়ার অধিকারকে প্রভাবিত করবে না।

আমি আরও ঘোষণা করছি যে আমি আমার চিকিৎসা ইতিহাস প্রদান করব, চিকিৎসা পরীক্ষা করাব, প্রয়োজনীয় ক্লিনিক্যাল উপাদান ও প্রাসঙ্গিক তথ্য প্রদান করব, প্রয়োজনীয় চিকিৎসা গ্রহণ করব এবং ফলো-আপ-এর জন্য আসব। আমি বুঝতে পেরেছি যে আমার পরিচয় প্রকাশ করা হবে না এবং প্রদত্ত সমস্ত তথ্য গোপন রাখা হবে।

অতএব, আমি এই গবেষণায় অংশগ্রহণ করতে সম্মত।

ঘোষণাকারীর নাম:
ঘোষণাকারীর স্বাক্ষর:
স্থান:
তারিখ:

গবেষক কর্মকর্তার নাম:
গবেষক কর্মকর্তার স্বাক্ষর:
স্থান:
তারিখ:

Participant Information Sheet

We invite you to take part in a research study titled:

"MatriSuraksha: An AI-Driven Referral Decision Support System for Safer Childbirth in West Bengal"

This study aims to leverage artificial intelligence (AI), Machine Learning (ML) and predictive analytics to support frontline healthcare providers by enabling early detection of maternal risk factors, stratification of case severity, and evidence-based referral decision-making.

Participation in this study is **entirely voluntary**. Before you decide, please take time to read this document carefully, discuss it with your family and friends, and ask any questions you may have. If you agree to participate, you will be asked to sign a consent form.

Purpose of the Research

The purpose of this study is to gather information to leverage artificial intelligence (AI), Machine Learning (ML) and predictive analytics to support frontline healthcare providers by enabling early detection of maternal risk factors, stratification of case severity, and evidence-based referral decision-making, which will help to develop targeted policy level decision by the public health authorities for safer pregnancy and childbirth events.

Procedures

No additional procedures will be performed that may cause discomfort or risk beyond routine clinical care. Only information about the disease condition and Geospatial-Epidemiological data will be collected.

Duration

This study will not affect your duration to diagnosis, treatment process, or the duration of your hospital stay.

Potential Benefits

In the long term, findings from this research may help improve intervention strategies for to develop targeted policy level decision by the public health authorities

Confidentiality

- All de-identified research records will be securely stored to the Principal Investigator.
- You will not be identified by name, address, or phone number in any research reports or publications.
- Records may include your initials, but the master list linking your identity to the study code will be kept in a locked file.
- While every effort will be made to maintain confidentiality, absolute confidentiality cannot be guaranteed.
- Research-related records may be reviewed by:
    - The Investigator and his research team

- The Institutional Ethics Review Board

Costs

- There are **no costs** to you for participating in this study.

Compensation

- You will not receive any payment or compensation for participation.

Voluntary Participation and Right to Withdraw

- Participation is voluntary.
- You are free to withdraw from the study at any time, without giving a reason.
- Choosing not to participate, or withdrawing later, will **not affect your right to receive medical care or benefits** to which you are otherwise entitled.
- The investigator may also withdraw you from the study, if necessary, even without your consent (e.g., for medical or administrative reasons).

Contact Information

If you have any questions, concerns, or complaints about this study—or if you believe you may have developed any problem related to participation—please contact:

Dr. Dilip Kumar Biswas, Deputy CMOH – III, Dakshin Dinajpur

Mobile No: 9748393477

Email: dilipbiswas29@gmail.com

प्रतिभागी सूचना पत्र (Participant Information Sheet)

हम आपको एक शोध अध्ययन में भाग लेने के लिए आमंत्रित कर रहे हैं जिसका शीर्षक है:

“मातृसुरक्षा: सुरक्षित प्रसव हेतु पश्चिम बंगाल में एआई-संचालित रेफरल निर्णय सहयोग प्रणाली”

यह अध्ययन कृत्रिम बुद्धिमत्ता (AI), मशीन लर्निंग (ML) और पूर्वानुमान विश्लेषण का उपयोग करके अग्रिम पंक्ति के स्वास्थ्यकर्मियों को सहयोग प्रदान करने का उद्देश्य रखता है, ताकि मातृ जोखिम कारकों की शीघ्र पहचान, मामलों की गंभीरता का वर्गीकरण, और साक्ष्य-आधारित रेफरल निर्णय-निर्धारण सक्षम हो सके।

इस अध्ययन में भाग लेना **पूरी तरह स्वैच्छिक** है। निर्णय लेने से पहले कृपया इस दस्तावेज़ को ध्यान से पढ़ें, अपने परिवार और मित्रों से चर्चा करें और यदि कोई प्रश्न हों तो हमारे स्टाफ से पूछें। यदि आप भाग लेने के लिए सहमत हैं, तो आपसे एक सहमति पत्र पर हस्ताक्षर करने के लिए कहा जाएगा।

अनुसंधान का उद्देश्य (Purpose of the Research)

इस अध्ययन का उद्देश्य जानकारी एकत्र करना है, ताकि कृत्रिम बुद्धिमत्ता (AI), मशीन लर्निंग (ML) और पूर्वानुमान विश्लेषण का उपयोग करके अग्रिम पंक्ति के स्वास्थ्यकर्मियों को सहयोग प्रदान किया जा सके। इसके माध्यम से मातृ जोखिम कारकों की शीघ्र पहचान, मामलों की गंभीरता का वर्गीकरण और साक्ष्य-आधारित रेफरल निर्णय-निर्धारण संभव होगा, जो सार्वजनिक स्वास्थ्य प्राधिकरणों को लक्षित नीतिगत निर्णय विकसित करने में मदद करेगा ताकि गर्भावस्था और प्रसव की प्रक्रिया अधिक सुरक्षित हो सके।

प्रक्रियाएँ (Procedures)

- कोई अतिरिक्त प्रक्रिया नहीं की जाएगी जो सामान्य चिकित्सा देखभाल से परे असुविधा या जोखिम पैदा करे।
- केवल रोग की स्थिति और भू-स्थानिक-रोग- महामारी विज्ञान संबंधी जानकारी एकत्र की जाएगी।

अवधि (Duration)

इस अध्ययन का आपके निदान, उपचार की प्रक्रिया या अस्पताल में भर्ती रहने की अवधि पर कोई प्रभाव नहीं पड़ेगा।

संभावित लाभ (Potential Benefits)

- दीर्घकाल में, इस शोध के निष्कर्ष सार्वजनिक स्वास्थ्य प्राधिकरणों द्वारा लक्षित नीतिगत निर्णय विकसित करने और हस्तक्षेप रणनीतियों को सुधारने में मदद कर सकते हैं।

गोपनीयता (Confidentiality)

- सभी **पहचान-रहित (de-identified)** शोध अभिलेख (records) को प्रधान अन्वेषक (Principal Investigator) द्वारा सुरक्षित रूप से संग्रहीत किया जाएगा।
- किसी भी रिपोर्ट या प्रकाशन में आपका नाम, पता या फोन नंबर उजागर नहीं किया जाएगा।

- रिकॉर्ड में आपके नाम के प्रथमाक्षर हो सकते हैं, लेकिन नाम और कोड संख्या की सूची एक सुरक्षित फाइल में रखी जाएगी।
- हर संभव प्रयास किया जाएगा कि आपकी पहचान गोपनीय रखी जाए, लेकिन पूर्ण गोपनीयता की गारंटी नहीं दी जा सकती।
- शोध-संबंधी रिकॉर्ड निम्नलिखित द्वारा देखे जा सकते हैं:
    - अन्वेषक और उनकी अनुसंधान टीम
    - संस्थागत आचार समिति (Institutional Ethics Review Board)

खर्च (Costs)

- इस अध्ययन में भाग लेने के लिए आपको कोई खर्च नहीं करना होगा।

प्रतिपूर्ति (Compensation)

- इस अध्ययन में भाग लेने के लिए आपको कोई भुगतान या प्रतिपूर्ति नहीं मिलेगी।

स्वैच्छिक भागीदारी और वापस लेने का अधिकार (Voluntary Participation and Right to Withdraw)

- इस अध्ययन में भागीदारी स्वैच्छिक है।
- आप किसी भी समय बिना कारण बताए अध्ययन से बाहर निकल सकते हैं।
- अध्ययन में भाग न लेने या बाद में वापस लेने से आपके चिकित्सा देखभाल पाने के अधिकार या अन्य लाभों पर कोई असर नहीं पड़ेगा।
- यदि आवश्यक हुआ तो अन्वेषक आपको आपकी अनुमति के बिना भी अध्ययन से बाहर कर सकते हैं (उदा. चिकित्सा या प्रशासनिक कारणों से)।

संपर्क जानकारी (Contact Information)

यदि इस अध्ययन के संबंध में आपके कोई प्रश्न, चिंताएँ या शिकायतें हों—या यदि आपको लगता है कि इस अध्ययन में भाग लेने से कोई समस्या उत्पन्न हुई है—तो कृपया संपर्क करें:

डॉ. दिलीप कुमार बिस्वास, उप मुख्य चिकित्सा अधिकारी (III), दक्षिण दिनाजपुर (प्रधान अन्वेषक / Principal Investigator)
मोबाइल नंबर: **9748393477**
ईमेल: dilipbiswas29@gmail.com

অংশগ্রহণকারী তথ্যপত্র (Participant Information Sheet)

আমরা আপনাকে একটি গবেষণা প্রকল্পে অংশগ্রহণ করার জন্য আমন্ত্রণ জানাচ্ছি, যার শিরোনাম:

“মাতৃসুরক্ষা: নিরাপদ প্রসবের জন্য পশ্চিমবঙ্গে এআই-চালিত রেফারেল সিদ্ধান্ত সহায়ক ব্যবস্থা”

এই গবেষণার উদ্দেশ্য হলো কৃত্রিম বুদ্ধিমত্তা (AI), মেশিন লার্নিং (ML) এবং ভবিষ্যদ্বাণীমূলক বিশ্লেষণ ব্যবহার করে সামনের সারির স্বাস্থ্যসেবাদাতাদের সহায়তা করা, যাতে মাতৃঝুঁকির কারণসমূহের প্রাথমিক সনাক্তকরণ, কেসের তীব্রতার শ্রেণীবিন্যাস এবং প্রমাণভিত্তিক রেফারেল সিদ্ধান্ত গ্রহণ সম্ভব হয়।

এই গবেষণায় অংশগ্রহণ **সম্পূর্ণ স্বেচ্ছামূলক** । সিদ্ধান্ত নেওয়ার আগে অনুগ্রহ করে এই নথিটি মনোযোগ দিয়ে পড়ুন, আপনার পরিবার ও বন্ধুদের সঙ্গে আলোচনা করুন এবং যেকোনো প্রশ্ন থাকলে আমাদের গবেষণা কর্মীদের জিজ্ঞাসা করুন। যদি আপনি অংশগ্রহণে সম্মত হন, তবে আপনাকে একটি সম্মতিপত্রে স্বাক্ষর করতে বলা হবে।

গবেষণার উদ্দেশ্য (Purpose of the Research)

এই গবেষণার উদ্দেশ্য হলো তথ্য সংগ্রহ করা, যাতে কৃত্রিম বুদ্ধিমত্তা (AI), মেশিন লার্নিং (ML) এবং ভবিষ্যদ্বাণীমূলক বিশ্লেষণের ব্যবহার করে সামনের সারির স্বাস্থ্যসেবাদাতাদের সহায়তা করা যায়। এর মাধ্যমে মাতৃঝুঁকির কারণসমূহের প্রাথমিক সনাক্তকরণ, কেসের তীব্রতার শ্রেণীবিন্যাস এবং প্রমাণভিত্তিক রেফারেল সিদ্ধান্ত গ্রহণ সম্ভব হবে, যা জনস্বাস্থ্য কর্তৃপক্ষকে লক্ষ্যভিত্তিক নীতিগত সিদ্ধান্ত গ্রহণে সহায়তা করবে, ফলে গর্ভাবস্থা ও প্রসব আরও নিরাপদ হবে।

পদ্ধতি (Procedures)

- এমন কোনো অতিরিক্ত প্রক্রিয়া করা হবে না যা নিয়মিত চিকিৎসা পরিষেবার বাইরে অস্বস্তি বা ঝুঁকি তৈরি করতে পারে।
- শুধুমাত্র রোগ সম্পর্কিত তথ্য এবং ভূ-স্থানিক-রোগতাত্ত্বিক তথ্য সংগ্রহ করা হবে।

সময়কাল (Duration)

এই গবেষণার কারণে আপনার রোগ নির্ণয়ের সময়, চিকিৎসা প্রক্রিয়া বা হাসপাতালে থাকার সময়কাল কোনোভাবেই প্রভাবিত হবে না।

সম্ভাব্য উপকারিতা (Potential Benefits)

- দীর্ঘমেয়াদে, এই গবেষণার ফলাফল জনস্বাস্থ্য কর্তৃপক্ষকে **লক্ষ্যভিত্তিক নীতিগত সিদ্ধান্ত** গ্রহণ ও হস্তক্ষেপ কৌশল উন্নত করতে সহায়তা করবে।

গোপনীয়তা (Confidentiality)

- সমস্ত **পরিচয়বিহীন (de-identified)** গবেষণা নথি প্রধান গবেষক (Principal Investigator)-এর কাছে নিরাপদে সংরক্ষিত থাকবে।
- কোনো গবেষণা প্রতিবেদন বা প্রকাশনায় আপনার নাম, ঠিকানা বা ফোন নম্বর প্রকাশ করা হবে না।
- রেকর্ডে আপনার নামের আদ্যক্ষর থাকতে পারে, তবে নাম ও কোড নম্বরের মূল তালিকা একটি তালাবদ্ধ ফাইলে রাখা হবে।
- আপনার পরিচয় গোপন রাখতে যথাসাধ্য চেষ্টা করা হবে, তবে **সম্পূর্ণ গোপনীয়তা নিশ্চিত করা যাবে না**।
- গবেষণা-সম্পর্কিত রেকর্ড নিম্নলিখিতদের দ্বারা দেখা যেতে পারে:
    - গবেষক ও তাঁর গবেষণা দল
    - প্রাতিষ্ঠানিক নৈতিক পর্যালোচনা বোর্ড (Institutional Ethics Review Board)

খরচ (Costs)

- এই গবেষণায় অংশগ্রহণের জন্য আপনার কোনো খরচ হবে না।

প্রতিদান (Compensation)

- এই গবেষণায় অংশগ্রহণের জন্য আপনি কোনো অর্থ বা প্রতিদান পাবেন না।

স্বেচ্ছাসেবী অংশগ্রহণ ও প্রত্যাহারের অধিকার (Voluntary Participation and Right to Withdraw)

- এই গবেষণায় অংশগ্রহণ সম্পূর্ণ স্বেচ্ছাসেবী।
- আপনি যে কোনো সময় কোনো কারণ না দেখিয়ে গবেষণা থেকে সরে আসতে পারেন।
- গবেষণায় অংশ না নিলে বা মাঝপথে সরে এলে আপনার চিকিৎসা পাওয়ার অধিকার বা অন্যান্য সুবিধার উপর কোনো প্রভাব পড়বে না।
- প্রয়োজনে গবেষক আপনার অনুমতি ছাড়াই আপনাকে গবেষণা থেকে সরিয়ে দিতে পারেন (যেমন চিকিৎসা বা প্রশাসনিক কারণে)

যোগাযোগের তথ্য (Contact Information)

এই গবেষণা সম্পর্কিত আপনার কোনো প্রশ্ন, উদ্বেগ বা অভিযোগ থাকলে—অথবা যদি মনে হয় এই গবেষণায় অংশগ্রহণের কারণে কোনো সমস্যা হয়েছে—তাহলে যোগাযোগ করুন:

ডা. দিলীপ কুমার বিশ্বাস, উপ মুখ্য চিকিৎসা আধিকারিক - III, দক্ষিণ দিনাজপুর (প্রধান গবেষক / Principal Investigator)
মোবাইল নম্বর: **9748393477**
ইমেল: dilipbiswas29@gmail.com

Draft Questionnaire for the Data Collection: Structured Case Record Forms

(This form will be finalized based on input from Pilot Study)

A. General and Demographic Details

1. Unique Study ID: _____

2. Date of Admission: ___/___/_____

3. PHC Name/Code: _____

4. Block: _____

5. Name of Pregnant Woman: _____

6. Age (in years): _____

7. Address: _____

8. Phone Number: _____

9. ANC Number: _____

10. Religion: [ ] Hindu [ ] Muslim [ ] Christian [ ] Others

11. Caste Category: [ ] General [ ] SC [ ] ST [ ] OBC

12. Education Level: [ ] Illiterate [ ] Primary [ ] Secondary [ ] Higher

13. Occupation: _____

14. Socioeconomic Status: [ ] BPL [ ] APL

15. Type of Family: [ ] Nuclear [ ] Joint

16. Distance of Residence from nearest PLHCU (in km): _____

B. Antenatal History

1. Gravida: _____

2. Parity: _____

3. Weeks of Gestation: _____

4. Number of ANC Visits: _____

5. Any History of Previous Complications: [ ] Yes [ ] No

- If Yes, Specify: _____

6. Hemoglobin (Hb) (g/dL): _____

7. Blood Pressure (mmHg): _____

8. Weight (kg): _____

9. Fundal Height (cm): _____

10. Fetal Heart Rate (FHR): _____

11. Urine Albumin: [ ] Negative [ ] Trace [ ] + [ ] ++

12. Blood Sugar (FBS/RBS/GTT): _____

13. Ultrasound Findings:

- Fetal Position: _____

- Placenta: _____

- AFI: _____

- Gestational Age: _____

14. Any Diagnosed Comorbidities:

- [ ] Anemia [ ] Gestational Diabetes [ ] PIH

- [ ] Hypothyroidism [ ] Other: _____

C. Intrapartum Details

1. Onset of Labour: Date ___/___/_____ Time: ____:____ AM/PM

2. Stage of Labour on Admission: [ ] Latent [ ] Active [ ] Second [ ] Third

3. Duration of Labour (in hours): _____

4. Maternal Vitals: Pulse: ___ BP: ___ Temp: ___

5. Vaginal Examination Findings:

- Cervical Dilatation: _____

- Effacement: _____

- Presenting Part: _____

- Station: _____

6. Fetal Heart Rate Monitoring: _____

7. Use of Partograph: [ ] Yes [ ] No

8. Any Interventions Done at PLHCU: [ ] IV Fluids [ ] Oxytocin [ ] Antibiotics [ ] Others: _____

D. Referral Details

1. Was the woman referred to a higher center? [ ] Yes [ ] No

- If Yes:

- Time of Referral Decision: ___:___ AM/PM

- Time of Departure: ___:___ AM/PM

- Name of Referral Facility: _____

- Mode of Transport: [ ] Ambulance [ ] Private [ ] Other

- Reason for Referral:

[ ] Prolonged Labour [ ] Fetal Distress [ ] PIH

[ ] Obstructed Labour [ ] APH/PPH [ ] Other: _____

- Was referral timely? [ ] Yes [ ] No

2. Details of Attending Healthcare Professional:

- Doctor with MBBS Qualification and less than one Year Working experience at PLHCU and regularly attending Antenatal and perinatal cases
- Doctor with MBBS Qualification and at least Year Working experience at PLHCU and regularly attending Antenatal and perinatal cases
- Doctor with MBBS Qualification and at least One Year House-staff ship in G&O
- Doctor with MBBS and DGO/MD/DNB (G&O) Qualification
- Nurses with ANM Training
- Nurses with GNM Training
- Nurses with BSc Training

E. Maternal Outcomes

1. Mode of Delivery: [ ] Normal [ ] Assisted [ ] Caesarean

2. Any Maternal Complications:

[ ] PPH [ ] Sepsis [ ] Eclampsia [ ] Retained Placenta

[ ] Maternal Death [ ] Other: _____

3. Date and Time of Delivery: ___/___/_____ ____:____

4. Length of Stay at PLHCU/Higher Center (in hours): _____

5. Place of Final Delivery with date and time:

F. Neonatal Outcomes

1. Sex of Baby: [ ] Male [ ] Female

2. Birth Weight (grams): _____

3. Need for Resuscitation? [ ] Yes [ ] No

3. Neonatal Status: [ ] Alive [ ] Stillbirth [ ] Death (within 7 or 42 days)

5. Referred to SNCU/NICU? [ ] Yes [ ] No

- If Yes, Reason: _____

G. Follow-up

1. Was postpartum follow-up done? [ ] Yes [ ] No

- If Yes, Days after delivery: ___

2. Postpartum Complications in Mother: [ ] Infection [ ] PPH [ ] High BP [ ] Other: _____

3. Neonatal Health on Follow-up: [ ] Normal [ ] Illness [ ] Hospitalized [ ] Death

H. Any Specific Comment by any healthcare provider related to the case

1. Name of the healthcare provider related to the case:

2. Designation of the healthcare provider related to the case:

3. **In your opinion, how usable and helpful is the Clinical Decision Support System (CDSS) for the healthcare professional at PLHCUs in assisting timely referrals and improving maternal and neonatal outcomes?**

- Very Useful and effective
- Useful and somewhat effective
- Not Useful and not effective

4. Any other relevant Comments: